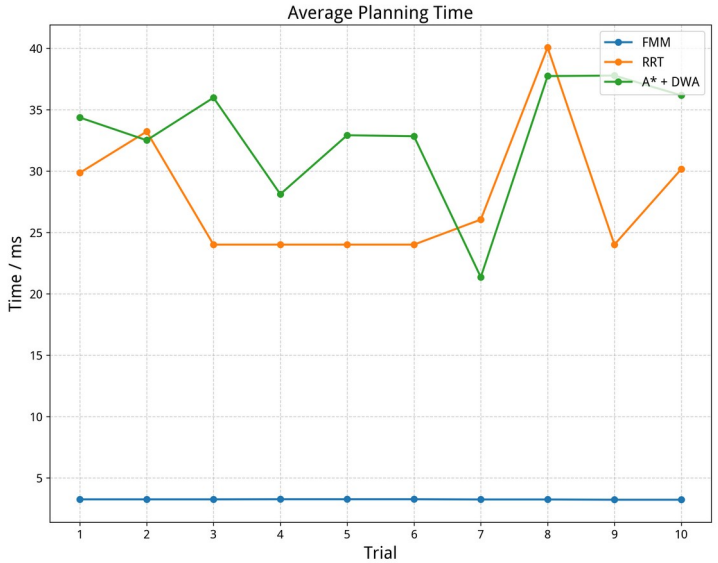
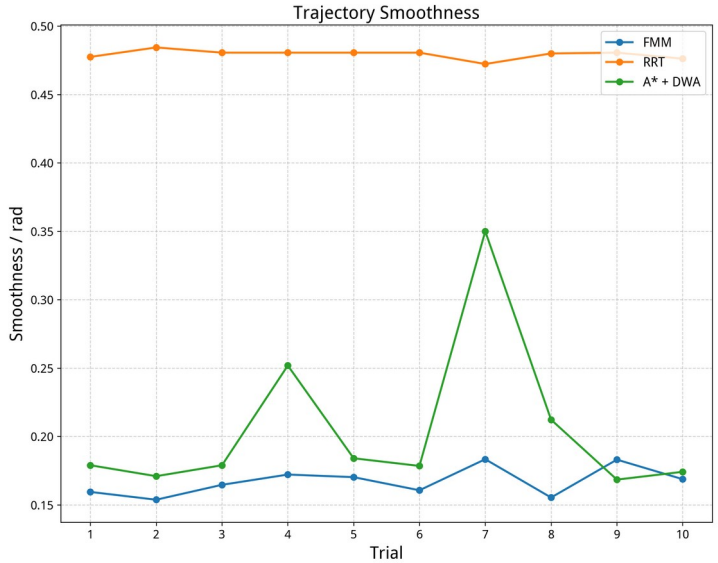
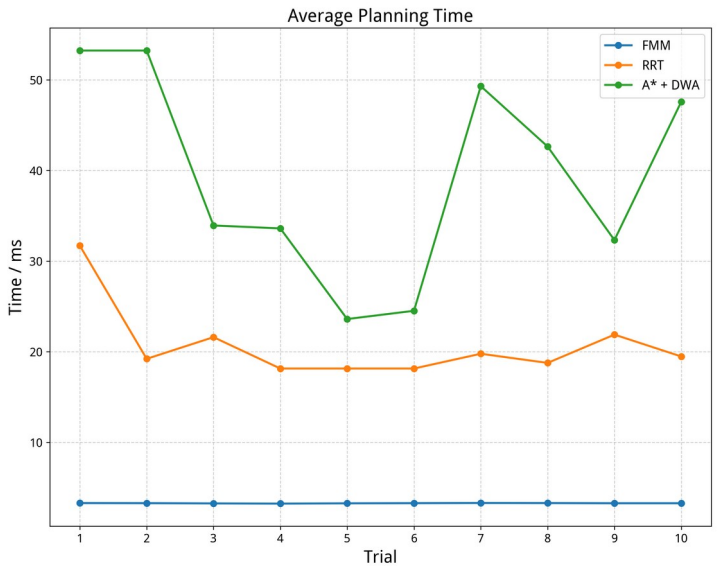

Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

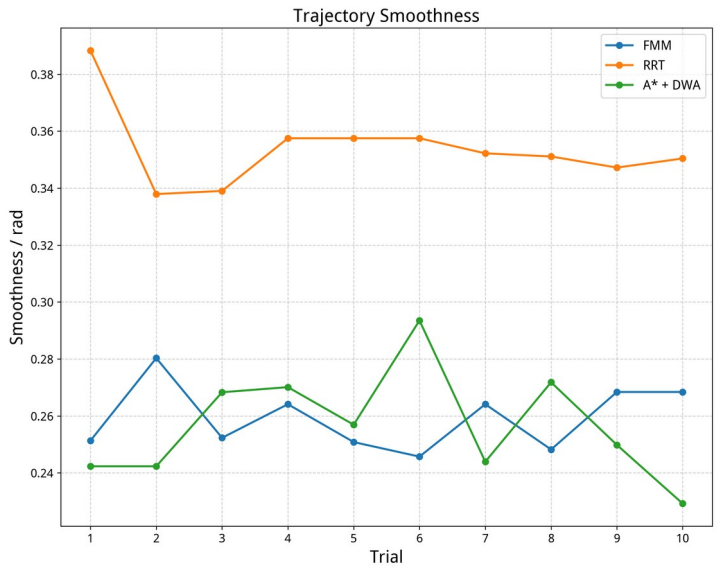
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

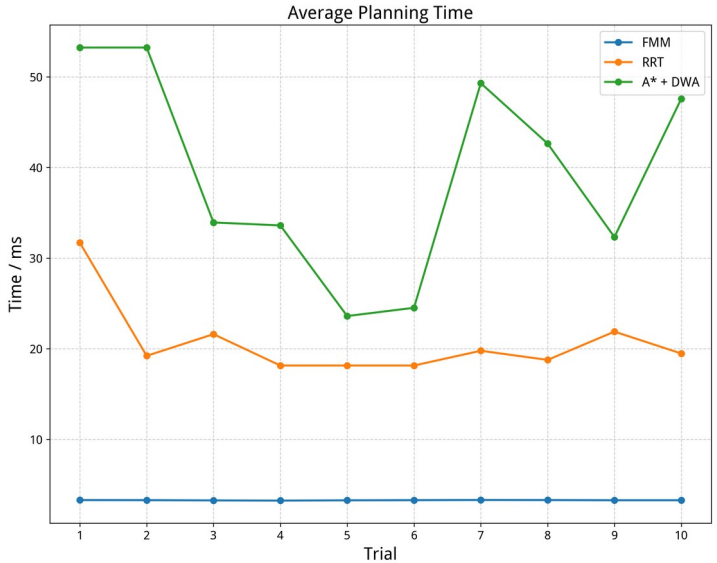
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

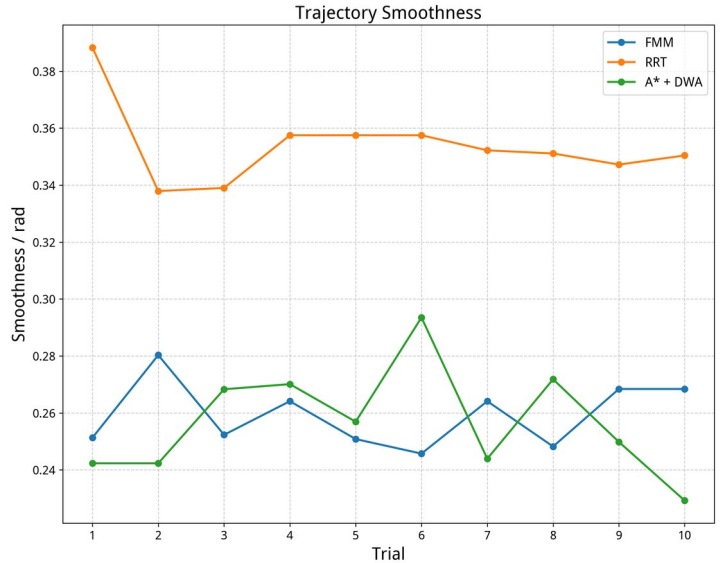
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

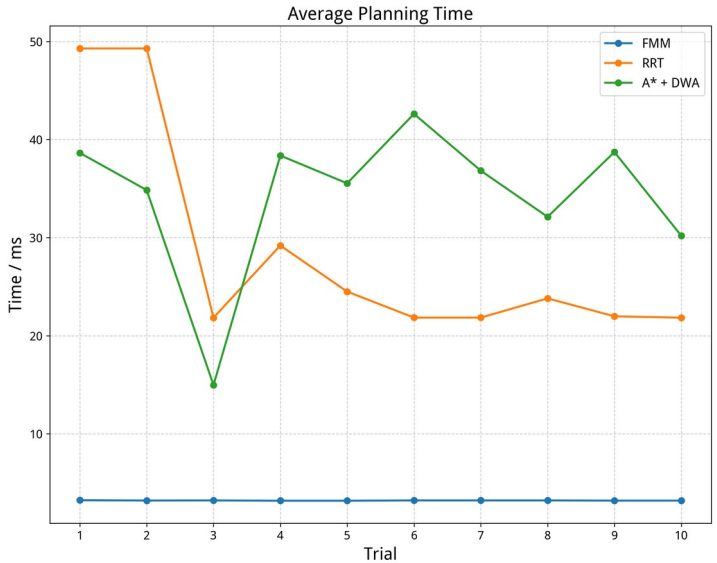
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

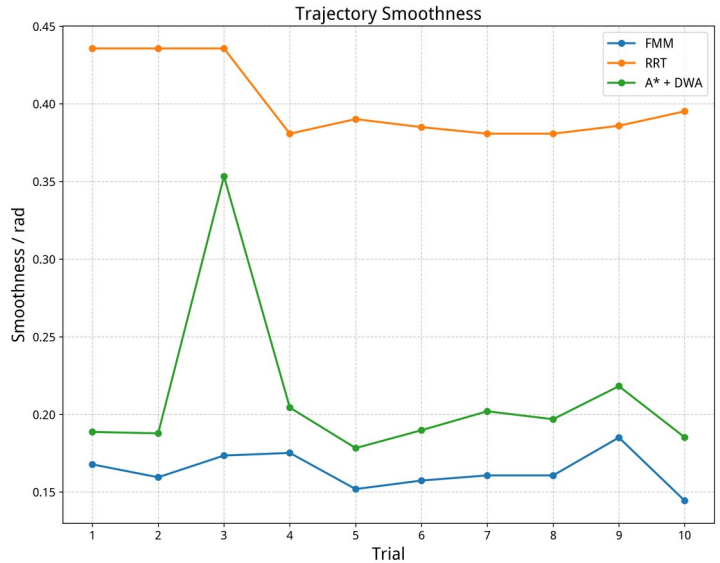
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

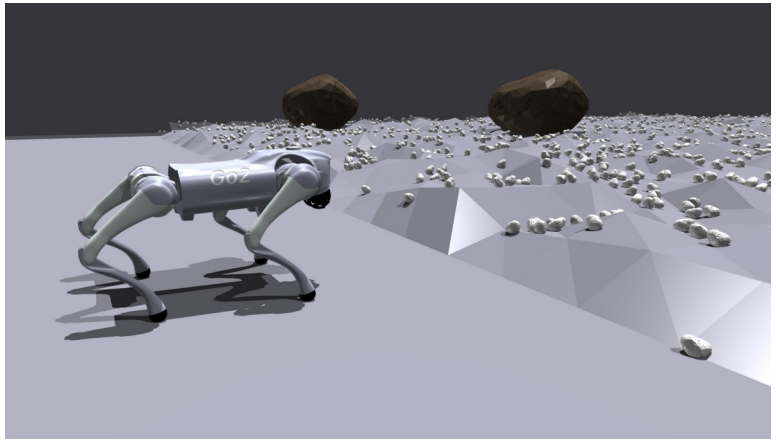
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

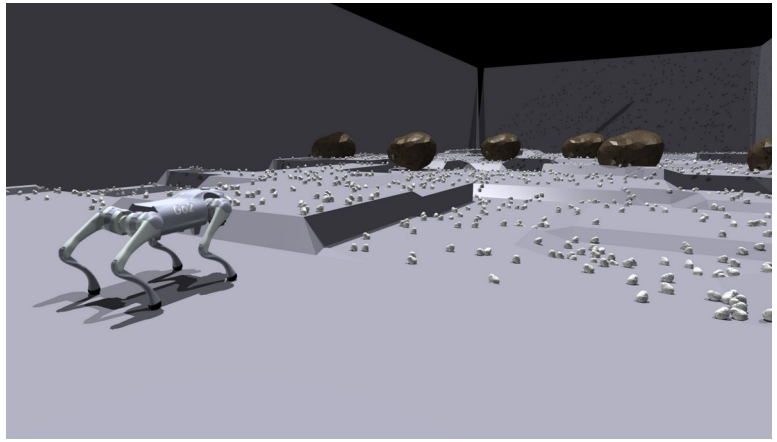
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

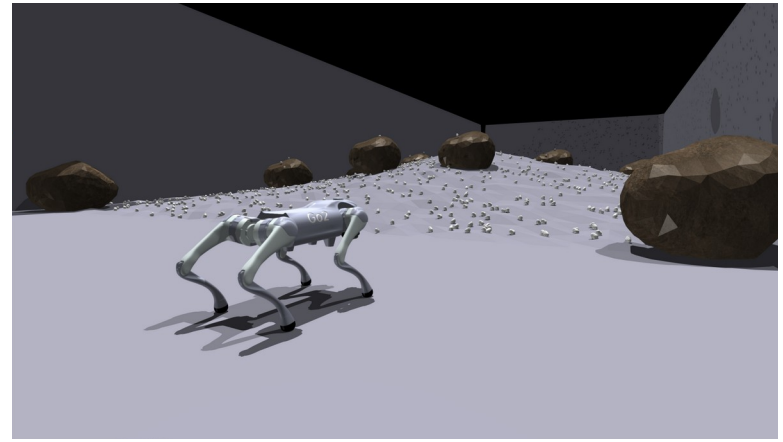
(b)



(a) Rough Slopes



(b) Discrete Obstacles



(c) Rough Flat





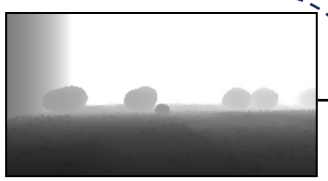
(a) Depth Image



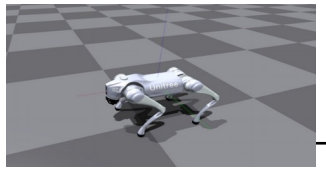
(b) BEV Image

Input

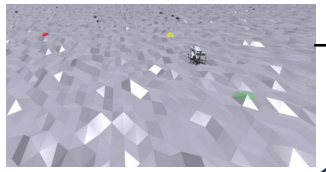
Depth Image



Quadruped Heading



Target Point Information



Navigation Controller

D-BEV Projection

Temporal Fusion

State Parsing

Trajectory Planning

FMM Planner

Bezier Optimization

HIM Training Pipeline

HIO

Representation Learning

Convergence Judgment

PPO

Policy Optimization

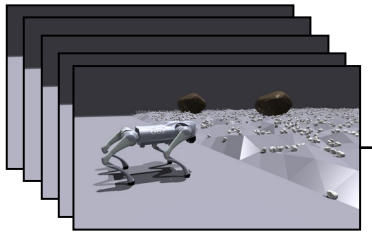
Converged

Network Initialization

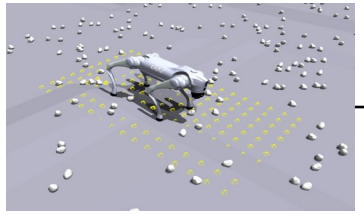
Optimal Model Saving

Input

Historical States



Terrain Observation



Gait Controller

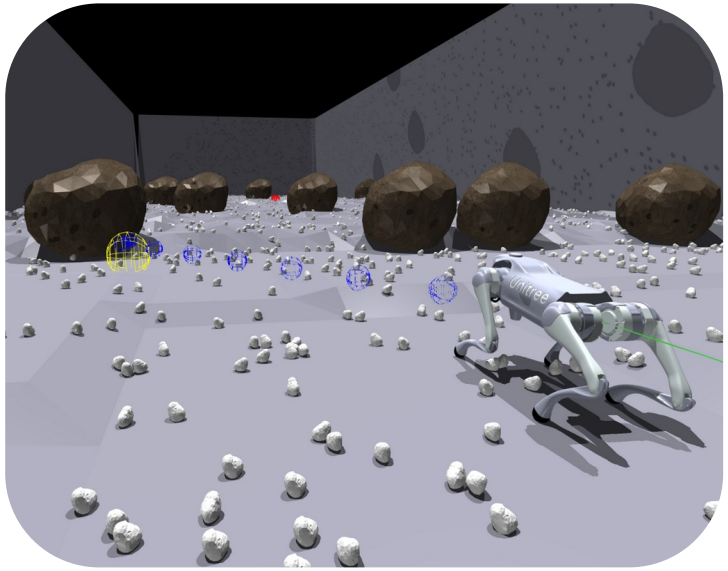
v_x, ω_{yaw}

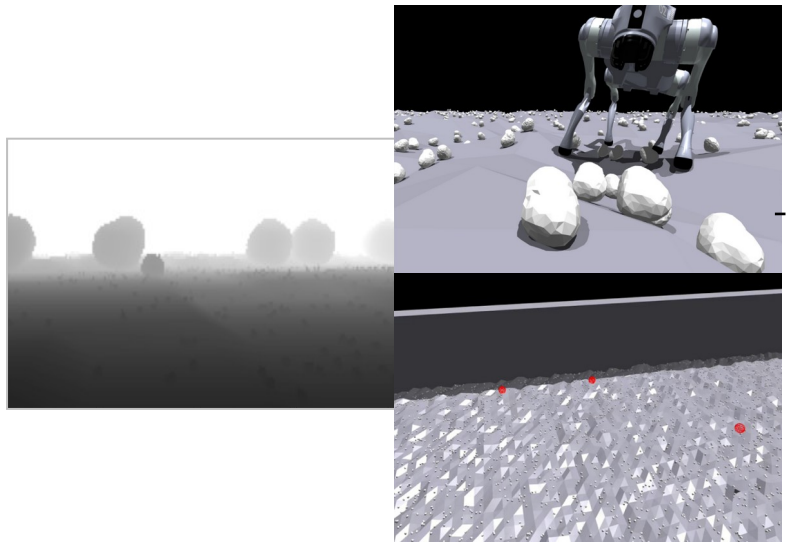
State Encoder

Current State

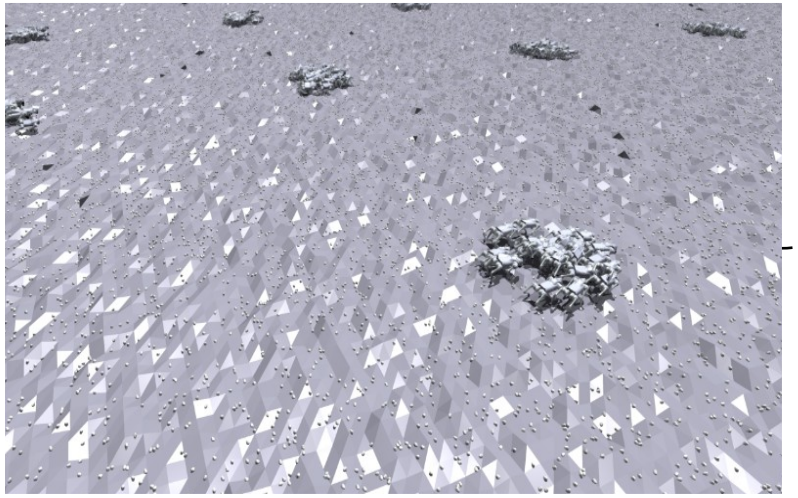
HIM-PPO Policy Network

Joint Position Commands





Depth Camera Perception Driving



Reward Function Driving

Trajectory Planning

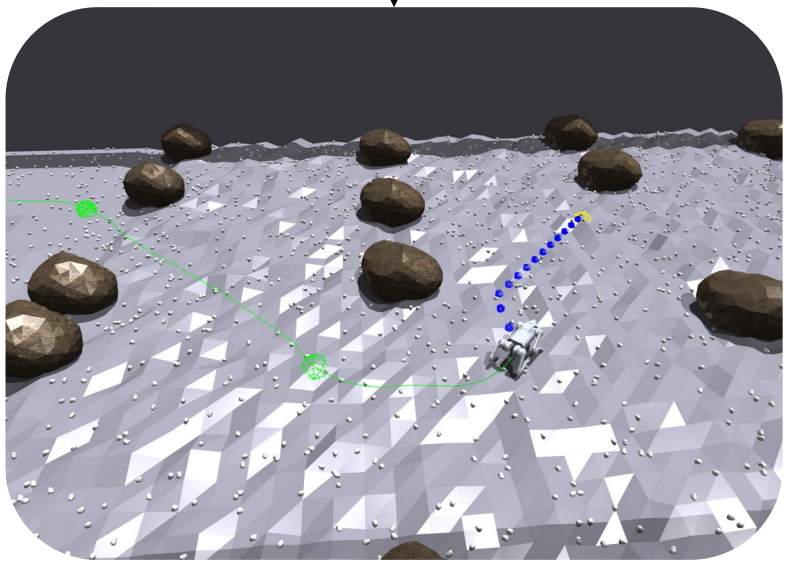
Navigation Controller

v_x, ω_{yaw}

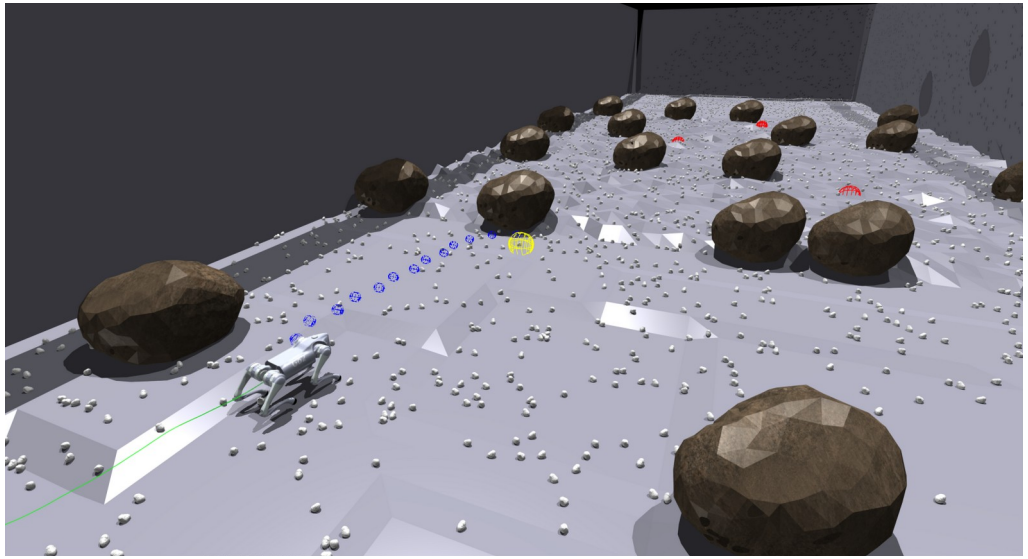
Gait Controller

Joint Position Commands

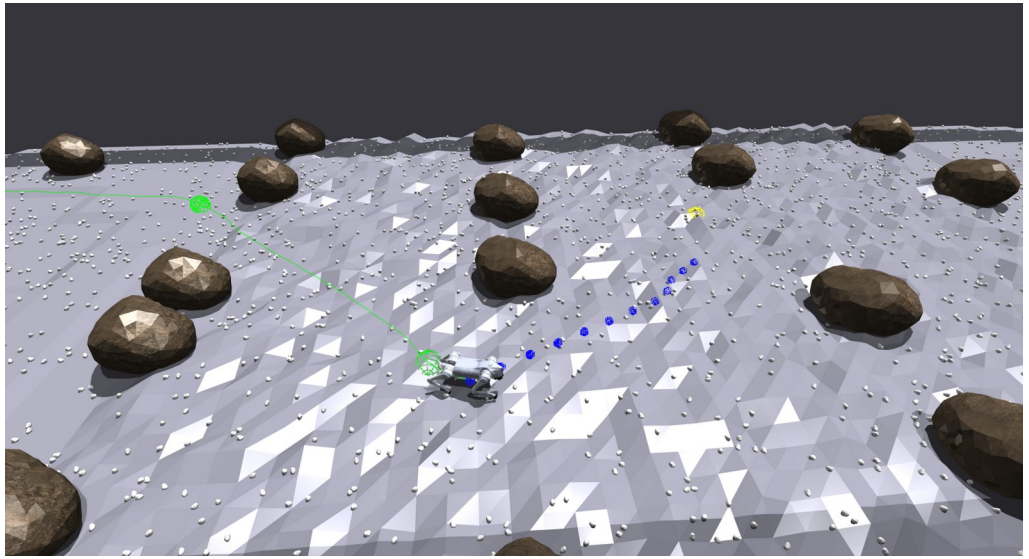
Reinforcement Learning



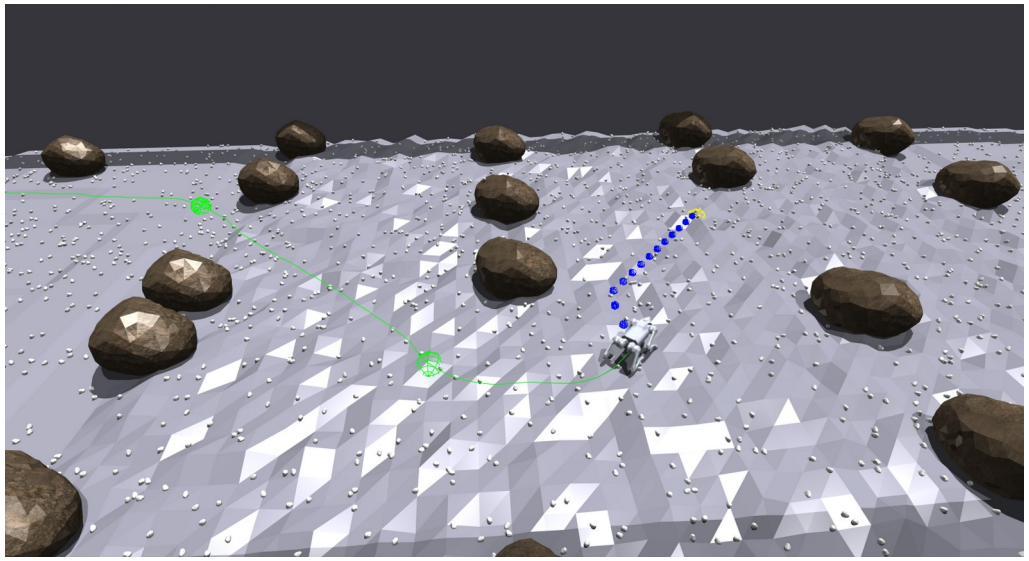
Autonomous Navigation



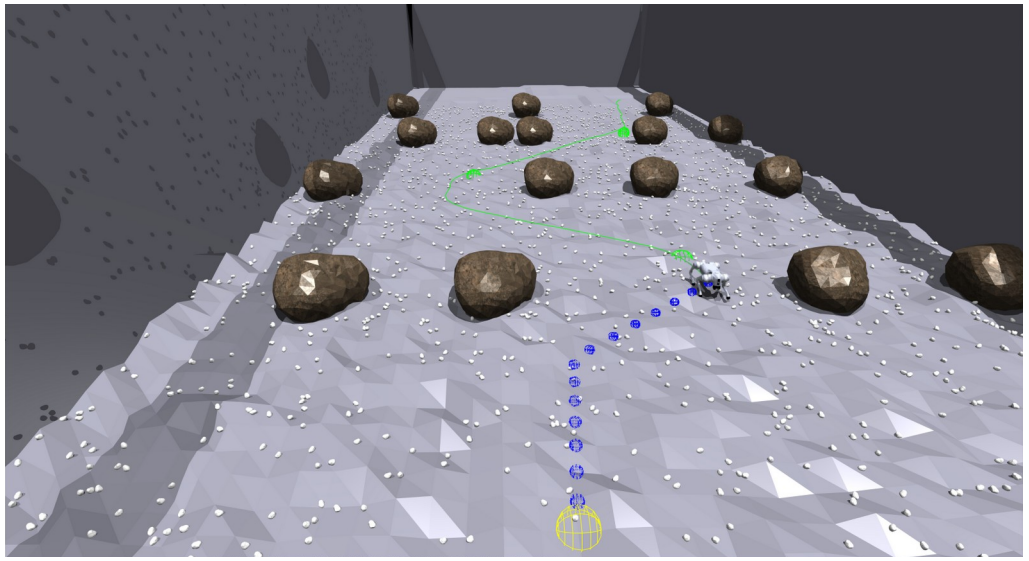
(a)Start Point — Waypoint A



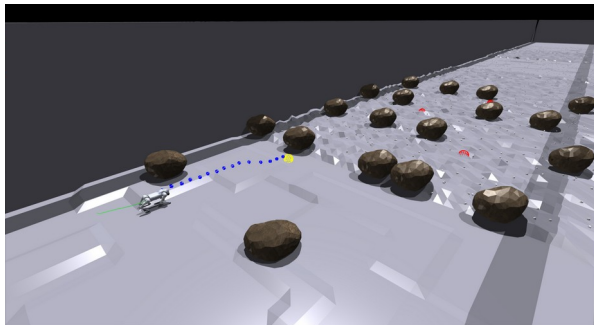
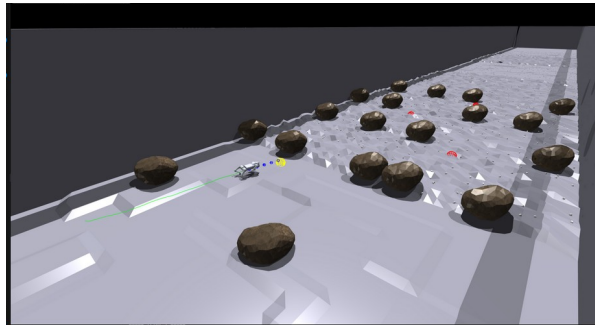
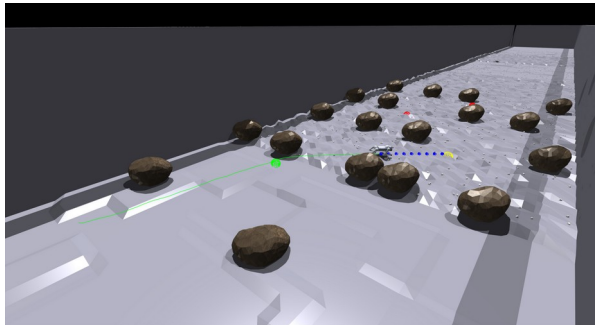
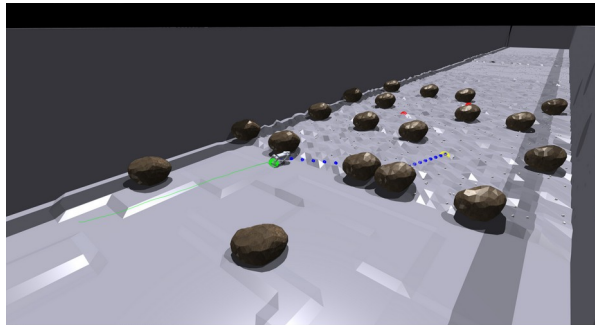
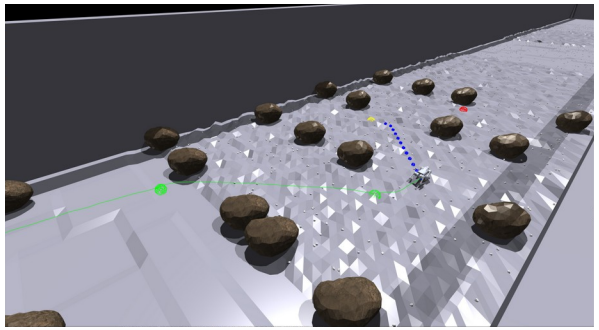
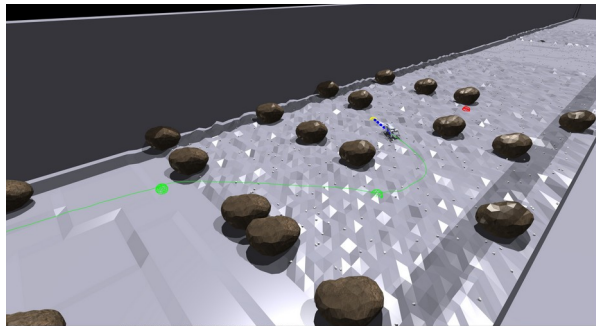
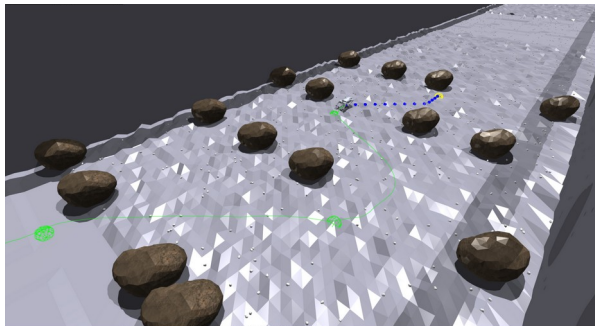
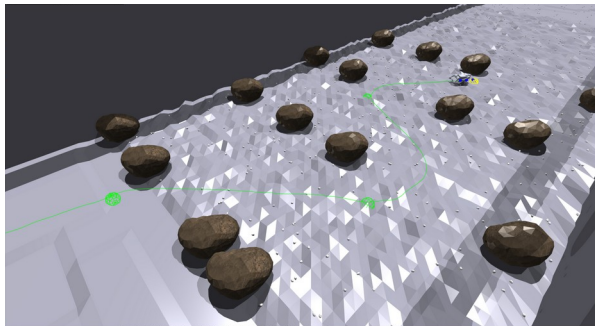
(b) Waypoint A — Waypoint B

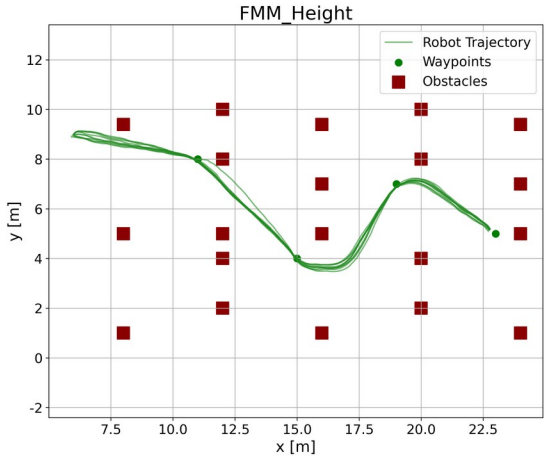


(c)Waypoint B — Waypoint C

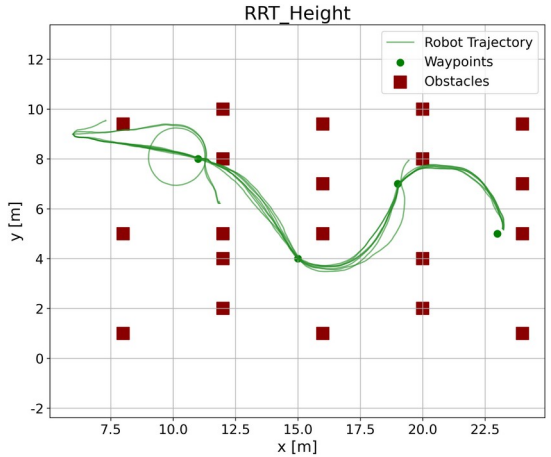


(d)Waypoint C — End Point

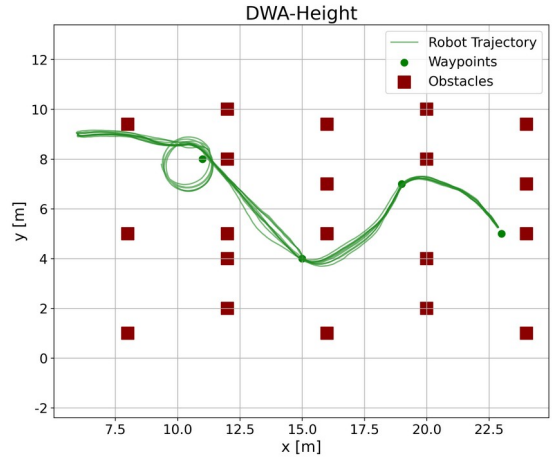










FMM_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

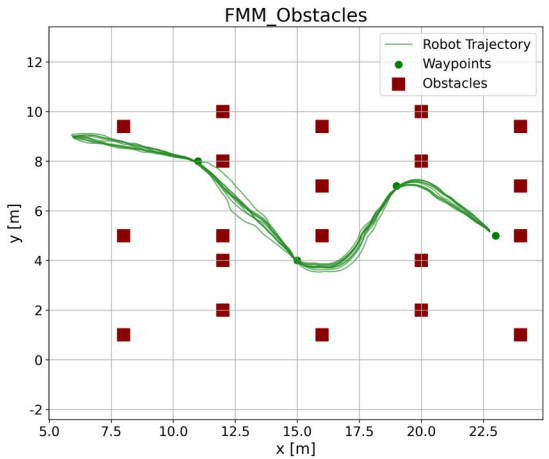
(a)


RRT_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

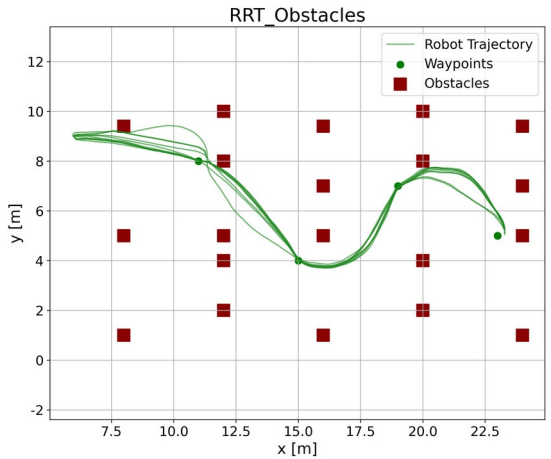
(b)


DWA-Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

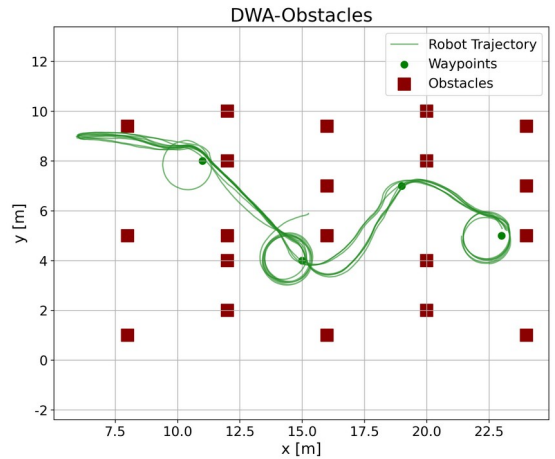
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

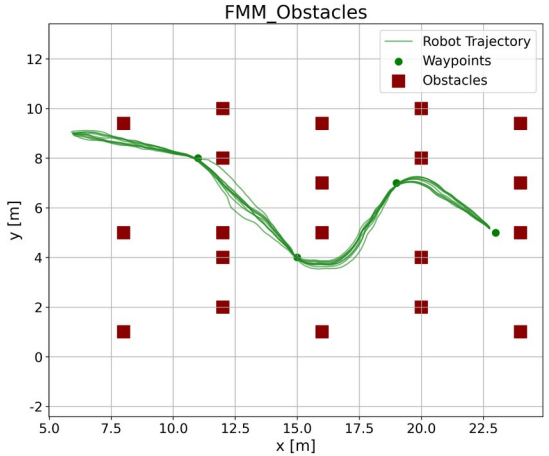
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

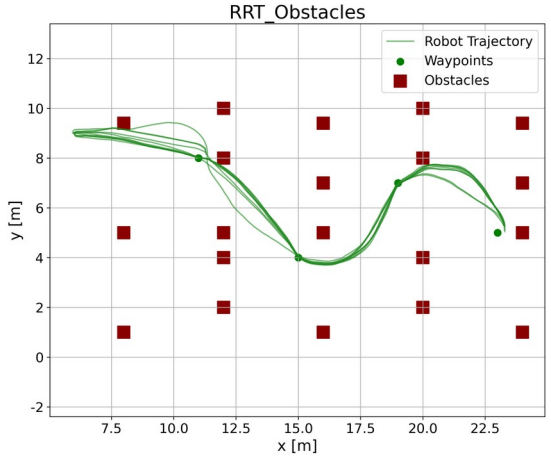
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

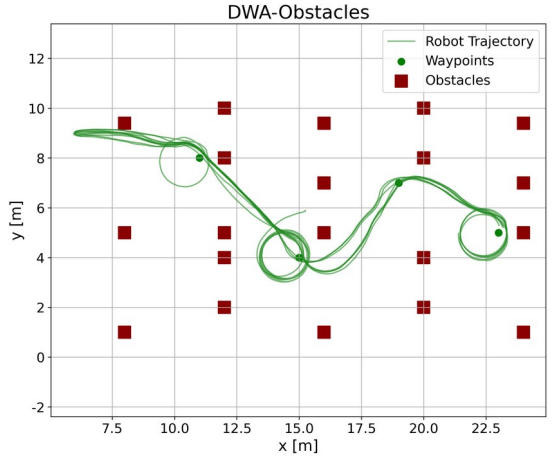
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

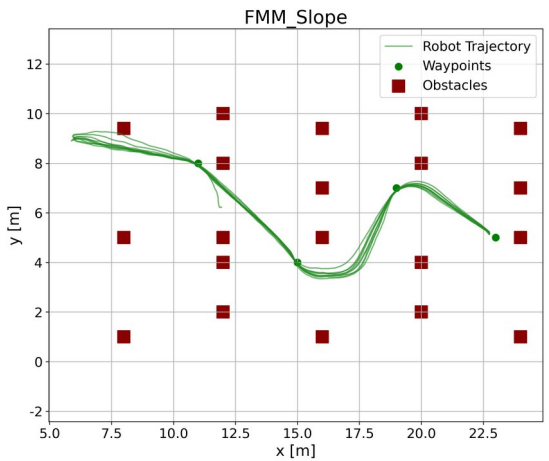
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

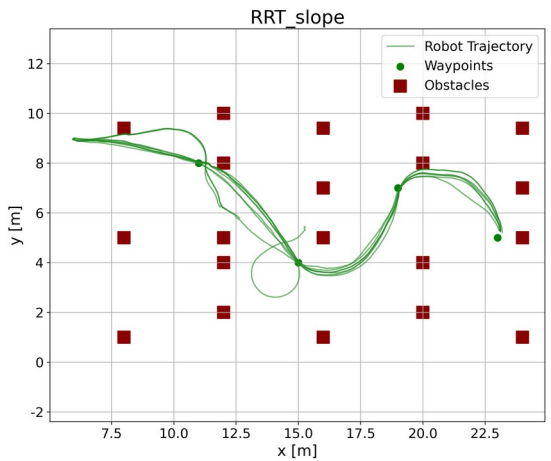
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

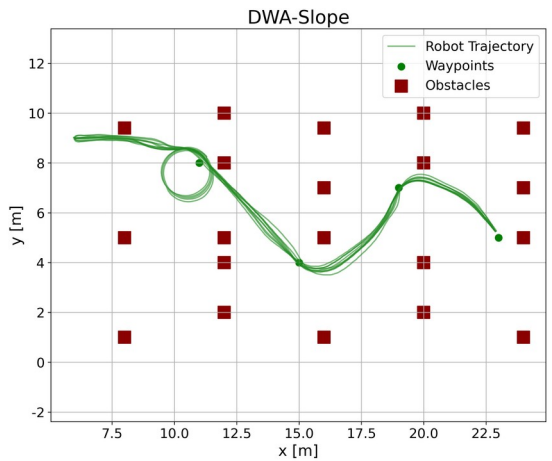
(c)


FMM_Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(a)


RRT_slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(b)


DWA-Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(c)